

Counting Arguments

CSU, Chico Math 350

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Rule of Product

If we repeat a trial, such as flipping a coin, many times, we can keep track of the possible combinations through a few propositions.

*When an experiment has two parts, If there are m ways to do the first part, and for each of those m ways there are n ways to do the second part, then there are $m * n$ ways to do both things.*

Rule of Product: Example

Let E_1 denote the selection of a rat from a cage containing one female (F) rat and one male (M) rat. Let E_2 denote the administering of either drug A , drug B , or a placebo to the selected rat.

- * How many possible outcomes are there?
- * What are the possible outcomes?

You can also illustrate the possibilities with a tree diagram, with paths diverging from each event's choices. The rule of product applies to cases with more than two events as well.

Exercise 1: Calendars

Each year starts on one of the seven days (Sunday through Saturday). Each year is either a leap year (i.e., it includes February 29) or not. How many different calendars are possible for a year?

Exercise 2: Chemical Reaction

A chemical engineer wishes to observe the effects of temperature, pressure, and catalyst concentration on the yield resulting from a certain reaction. If they intend to include two different temperatures, three pressures, and two levels of catalyst, how many different runs must they make in order to observe each temperature-pressure catalyst combination exactly twice?

Exercise 3: Restaurant

A restaurant offers a choice of four appetizers, fourteen entrees, and six desserts. How many different meals are possible if a diner intends to order only three items, one from each menu? That is, you can't have two desserts and no entree.

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We've already seen problems where we pick a handful of choices from a larger sample without replacement. There's a formula to calculate how many combinations exist in these cases:

The number of ways of choosing k distinct objects from a set of n is given by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

where

$$n! = 1 * 2 * \dots * n; 0! = 1$$

RStudio uses the function `choose(n,k)` to perform this calculation.

Exercise 4: Team Composition

A competitive sport club consisting of twenty-five members is trying to assign a total of nine slots to its team for an upcoming game. Among the members, ten are experienced and fifteen are novices. The team requires a two-to-one ratio of experienced players to novices—how many ways can the club choose the positions?

Recall not just the formulation for combinations, but also the Rule of Product here.

Exercise 5: Internships

Nine students, five statistics majors and 4 computer science majors, interview for four summer internships sponsored by a data analysis company.

1. In how many ways can the company choose a set of four interns?
2. In how many can the company choose 2 stat majors and 2 computer science majors?
3. How many sets of four can be picked such that not everyone in the set is the same major?

Hint for 3: Find the way that groups of 4 can be picked such that EVERYONE is in the same major, and subtract that from the total number of ways to choose 4 interns.

Example: Gender Equality in Promotions

Ten equally qualified marketing assistants are candidates for promotion to associate buyer; seven are men and three are women. If the company intends to promote four of the ten at random, what is the probability that exactly two of the four are women?

```
total <- ___  
two_women <- ___  
(prob_two_women <- ___ )
```

Exercise 6: Return of the Urn

An urn contains twenty chips, numbered 1 through 20. Two are drawn simultaneously. What is the **probability** that the numbers of the two chips will differ by more than 2?

Hint: Calculate the complement (how many ways can the numbers differ by 2 or LESS?) and subtract from one.

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.